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**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING**

**Academic Year: 2022-23**

Programme: B.Tech / MBA Tech (All Stream) / B.Tech Integrated (Civil) Year: II/III/IV Semester:III/V/VII

Subject: Management Accounting for Engineers

Date: 25 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 05

**Final Examination (2022-23)/ Re-Examination (2021-22)**

**Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.**

1) Question No. _1___ is compulsory..
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

| **Q.1** | | **Answer briefly and show detailed calculations.** | 20 |
|---|---|---|---|
| CO- 1; SO- ; BL- Remember | a | What are the essentials of a Good Costing System? | 5 |
| CO-4 ; SO- ; BL- Evaluate | b | Radhika Pvt. Ltd. Produces a product "R". The unit selling price and variable cost per unit is Rs.1000 and Rs.750 respectively. Fixed cost is Rs.25,50,000 p.a.<br>a. Calculate Break-even point of the company.<br>b. Also, determine the expected sales if desired profit is Rs.7,50,000. | 5 |
| CO- 4 ; SO- ; BL- Evaluate | c | From the following information compute, Material Cost Variances, Material Price Variance and Material Usage Variance. | 5 |
| CO- 2 ; SO- ; BL- Apply | d | The annual demand for a product is 6,400 units. The unit cost is ₹ 6 and inventory-carrying cost per unit per annum is 25% of the average inventory cost. If the cost of procurement is ₹ 75. Calculate:<br>(a) Economic order quantity,<br>(b) Number of orders per annum, and<br>(c) Time between two consecutive orders. | 5 |

| Material | Standard cost for 1 unit | | | Actual cost for 10 units | | |
|---|---|---|---|---|---|---|
| | Quantity | Rate | Amount | Quantity | Rate | Amount |
| X | 60 | 15 | 900 | 640 | 17.5 | 11200 |
| Y | 80 | 20 | 1600 | 950 | 18 | 17100 |
| Z | 100 | 25 | 2500 | 870 | 27.5 | 23925 |

| | | | |
|---|---|---|---|
| **Q.2**<br>CO-2 ; SO-;<br>BL- Apply | a | From the following details furnished by Mittal Ltd. for the year ended 31st March 2022, prepare a Cost Sheet showing the elements of cost also find out the profit for the year.<br><br>**Particulars** / **Rs.**<br>Salesman Salary — 45000<br>Opening stock of Finished goods (2000 units) — 560000<br>Directors fees — 97800<br>Indirect wages — 345000<br>Repairs to office furniture — 40500<br>Works Managers salary — 109700<br>Showroom expenses — 11300<br>Depreciation on Computer — 1200<br>Indirect Materials — 630000<br>Depreciation on Plant and Machinery — 27500<br>Advertisements — 153000<br>Office Salary — 790000<br>Direct wages — 20200<br>Direct materials — 97500<br>Direct expenses — 32000<br>Closing stock of Finished goods (2000 units) — ----------<br><br>Other information :<br>1. Closing stock of Finished goods is to be valued at cost of production<br>2. Profit desired on sales is 20%<br>3. Number of units sold during the year was 10000 | 15 |
| **Q.2**<br>CO-2; SO-;<br>BL- Understand | b | Explain the benefits of good inventory control system for an organization. | 5 |

| | | | |
|---|---|---|---|
| **Q.3**<br>CO-2; SO-;<br>BL-Apply | a | For the manufacture of a product X, two components A and B are used.<br>The following particulars about the components are:<br>Normal usage (per week) 300 nos.<br>Maximum usage (per week) 450 nos.<br>Minimum usage (per week) 150 nos.<br>Reorder quantity A-2400 nos. , B-3600 nos.<br>Reorder period A-4 to 6 weeks, B- 2 to 4 weeks<br>You are required to calculate the following for both the components.<br>a) Reorder quantity<br>b) Maximum level and<br>c) Minimum level. | 8 |
| **Q.3**<br>CO-3 SO-;<br>BL-<br>Understand | b | Given are the information of XYZ Ltd.<br>Particulars / X / Y<br>Raw material per unit / 160 / 120<br>Wages / 80 / 60<br>Selling price per unit / 500 / 400<br>Fixed cost is Rs.5,00,000 and variable overheads are 150% of wages.<br>Alternative sales mix:<br>a. 3500 unit of X and 2500 units of Y.<br>b. 6000 units of X and Nil units of Y.<br>c. 4000 unit of X and 2000 units of Y.<br>From the above information, you are required to find out the most profitable sales mix. | 12 |
| **Q.4**<br>CO- 3; SO-;<br>BL-Apply | a | A factory is currently running at 60% capacity and produces 6000 units The cost per unit details is given below:<br>Material Rs.15<br>Labour Rs. 4<br>Factory overheads Rs. 15 (Rs. 6 fixed )<br>Administrative overheads Rs.5 (Rs. 3 fixed )<br>The current selling price is Rs. 50 per unit. At 70% working, material cost per unit increases by 5% and selling price per unit falls by 2%. At 90% working, material cost per unit increases by 8% and selling price per unit falls by 5%. Prepare a flexible budget showing profits of the factory at 70% and 90% working and offer your comments. | 15 |

| | | | |
|---|---|---|---|
| **Q.4**<br>CO- 4; SO-;<br>BL- Evaluate | b | Explain FIVE applications of Marginal costing for taking various types of managerial decisions. | 05 |
| **Q.5**<br>CO-5; SO-;<br>BL-Apply | a | Ambuja Garments produces two products "SUCHI" and "RUCHI" which are popular in market. The management has four different options to produce both the products, given as under.<br><br>**Particulars** / **Suchi (Units)** / **Ruchi (Units)**<br>Option I: 800 / 600<br>Option II: 1,600 / --<br>Option III: -- / 1,300<br>Option IV: 1,100 / 500<br><br>**The per unit production cost/sale data are :**<br><br>**Particulars** / **Suchi** / **Ruchi**<br>Direct Material (₹): 25 / 30<br>Direct Labour (Hours): 10 / 12<br>Variable Factory Overheads is 100% of direct Labour cost for both products<br>Selling Price (₹): 75 / 90<br>Labour Rate is ₹ 2 per hour<br>Total Fixed Overheads for both products ₹ 10,000<br><br>You are required to evaluate the options and identify the most profitable sales-mix. | 15 |
| **Q.5**<br>CO-2; SO-;<br>BL- Evaluate | b | The standard material cost for 100 kg of a product Coke is made up of :<br>Material A- 30 kg @ Rs 4 per kg<br>Material B- 40 kg @ Rs.5 per kg<br>Material C- 80 kg @ Rs. 6 per kg<br>A batch of 500 kg of product Coke is made up of :<br>Material A – 140 kg at a total cost of Rs. 588<br>Material B- 220 kg at a total cost of Rs. 1056<br>Material C- 440 kg at a total cost of Rs. 2860<br>Calculate all Material Variances. | 5 |

<table>
<tr>
<td>Q.6<br>CO- 4, SO-;<br>BL-<br>Evaluate</td>
<td>a</td>
<td>Following information is available in respect of G Ltd. And D Ltd:<br><br>
<table>
<tr><th>Particulars</th><th>G Ltd. (Rs )</th><th>D Ltd. (Rs)</th></tr>
<tr><td>Sales</td><td>11,00,000</td><td>14,00,000</td></tr>
<tr><td>Variable cost</td><td>8,80,000</td><td>10,50,000</td></tr>
<tr><td>Profit</td><td>1,20,000</td><td>2,00,000</td></tr>
</table>
<br>Calculate:<br>
i P/V Ratio of both companies<br>
ii. Fixed cost of both companies<br>
iii. Break-even point of both companies<br>
iv. Sales to earn profit of Rs 2,10,000 by each company<br>
v. Margin of safety of 'D' Ltd.</td>
<td>15</td>
</tr>
<tr>
<td>Q.6<br>CO- 1; SO-;<br>BL-<br>Remember</td>
<td>b</td>
<td>Explain any FIVE limitations of Financial Accounting.</td>
<td>5</td>
</tr>
<tr>
<td>Q.7<br>CO- 1; SO-;<br>BL-<br>Remember</td>
<td>a</td>
<td>Write Short Notes on:<br><br>1. Management Accounting<br>2. Activity Based Costing (ABC)<br>3. Importance of Budgeting</td>
<td>15</td>
</tr>
<tr>
<td>Q.7<br>CO- 4; SO-;<br>BL- Evaluate</td>
<td>b</td>
<td>Using the following information, calculate labour cost, rate , efficiency variances :<br>
The budgeted labour force for producing product X is :<br>
20 Semi-skilled workers @ 75 paise per hour for 50 hours<br>
10 Skilled workers @ Rs 1.25 per hour for 50 hours<br>
The actual labour force employed for producing X is :<br>
22 Semi-skilled workers @ 80 paisa per hour for 50 hours<br>
8 Skilled workers @ Rs 1.20 per hour for 50 hours</td>
<td>5</td>
</tr>
</table>

**THE END**